

JORDAN RIVER MINES LTD — SUNRO MINE

*NORTHGATE*

## MONTHLY REPORT

PROPERTY FILE 92C073(8E)

GEOLOGY DEPARTMENT

G. MEUSY — @ Mine

DATE : JUL 1973

The access ramp to the Cave Zone was brought up to 5425 Level and broken through into the Alimak Raise. The last rounds exposed mineralization of the Cave A.

The last lift in the River C Shrinkage Stope was completed and breasting was discontinued due to ground support problems, creating excessive dilution.

Production went on in the Cave A longhole stope with slightly better results.

The exploration drilling program was completed and established the discontinuity of the Cave B towards the north west. Definition drilling has been resumed in the Cave A Zone on 5400, from the upramp.

### I - RAMPING

Map 1

79 feet were driven, resulting in the breakthrough in the Alimak raise on 5425 level. The tectonic features exposed include the eastern limb of #1 Branch West, a fault striking north and dipping 15 degrees east and the Cave Fault. Secondary faults and fractures between the major accidents account for a fairly shattered ground. Mineralization was mapped past the Cave Fault in two main shoots, as it was expected.

The eastern limb, probably connected to the main body to the north, is fairly high in Pyrrhotite stringers and veinlets (general trend N 20 degrees W) with little disseminated Chalco-pyrite in specks and tension fractures filling. Chip sampling on the walls returned 20' at 1.00% Cu (northern wall) and 15' at .79% Cu (southern wall).

The western limb seems richer in copper and the face is still in it.

Diamond drilling in progress on Section 1100 will clarify the relations between these two oreshoots to the north of the ramp,

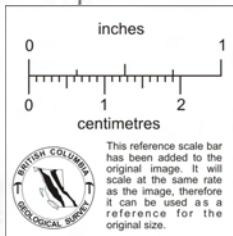
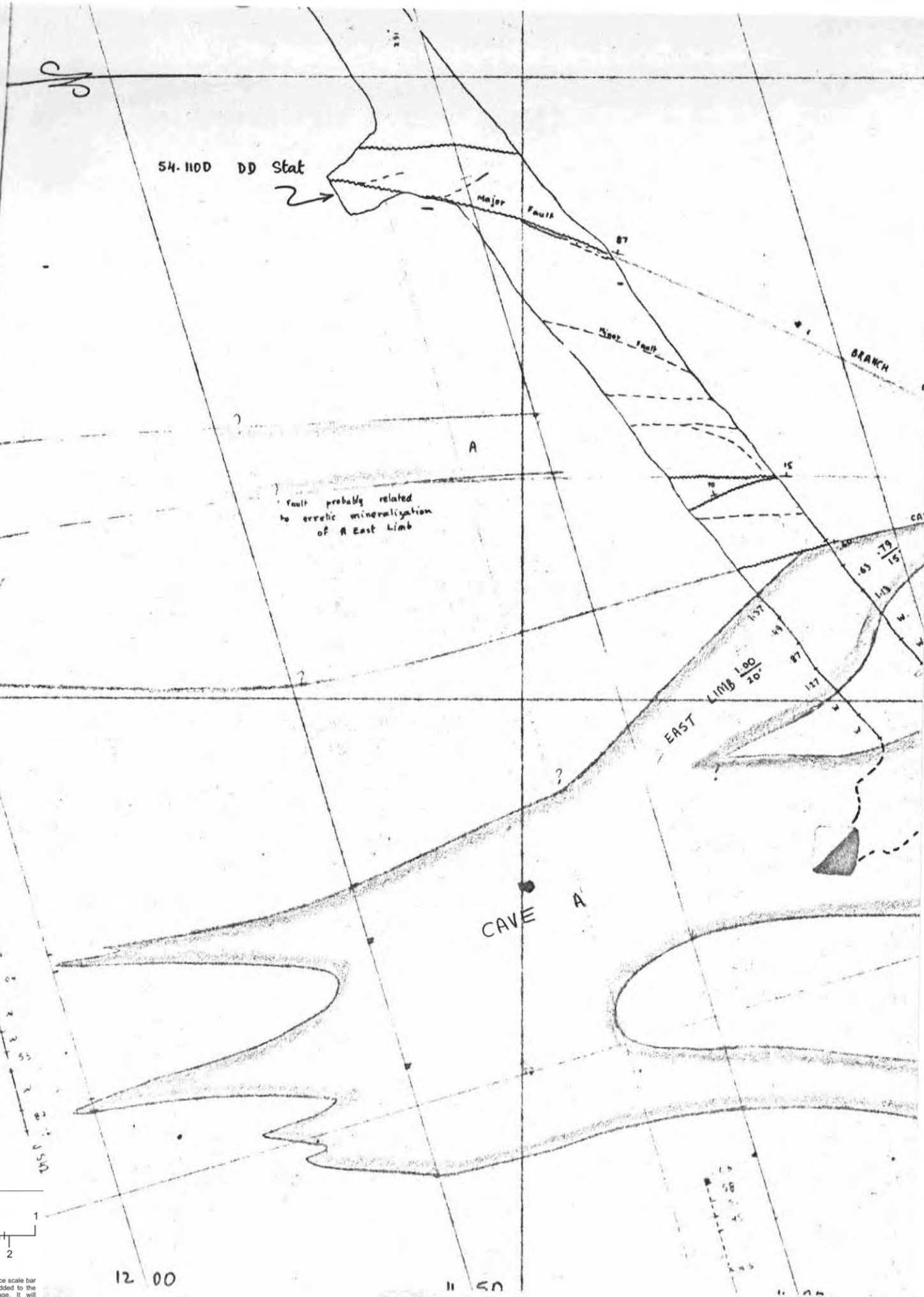
### II - RAISING

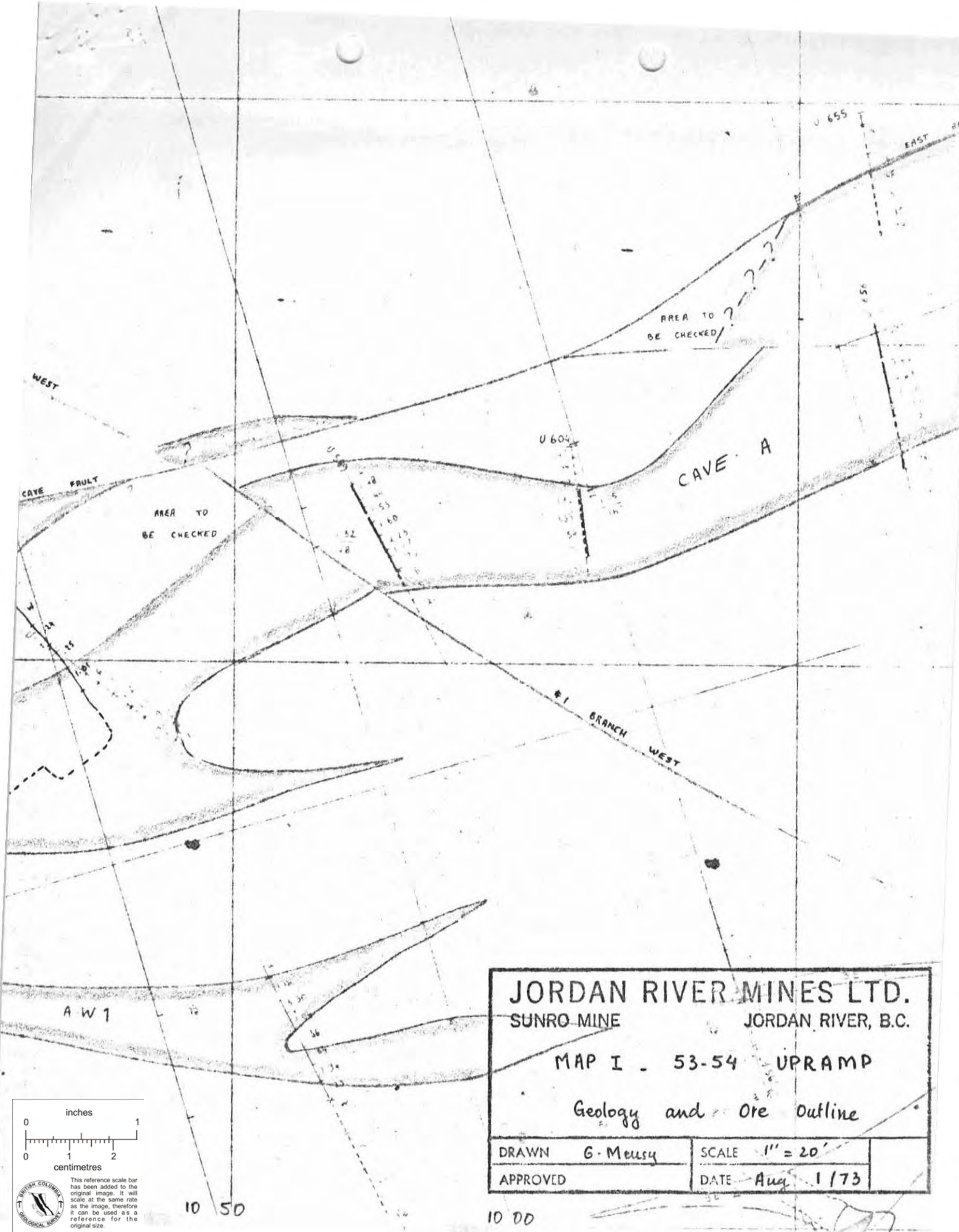
Raising in the Cave A South from 52 to 53 was resumed at the end of the month, and 6 feet were driven.

### III - STOPING

#### III - A - CAVE A STOPE

Longhole drilling and blasting went on in the Cave A Stope (southern part). A total of 20,263 Tons were pulled and muck sampling averages .83% Cu to July 26th (last samples not assayed to date).





# JORDAN RIVER MINES LTD.

SUNRO MINE

JORDAN RIVER, B.C.

MAP I - 53-54 UPRAMP

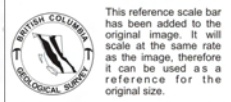
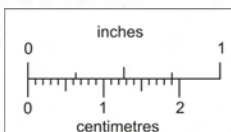
Geology and Ore Outline

DRAWN G. Meusy

SCALE 1" = 20'

APPROVED

DATE Aug 1/73



This reference scale bar has been added to the original image. It will scale at the same rate as the image, therefore it can be used as a reference for the original size.



91 Grade forecast allowing for 10% dilution was .95 Cu. Total dilution (actual) is therefore about 22%.

III - B - CAVE B STOPE

1,808 Tons were drawn from this stope, for which muck sampling indicates .87% Cu (Sampling efficiency equals 42%). Grade estimate for this muck was .85% Cu.

III - C - RIVER C STOPE

MAP III - C

4,574 Tons were broken in the shrinkage stope this month. (Average grade 1.75% Cu.)

A fairly heavy dilution was encountered which in addition to ground support problems, led to the abandon of this stope.

The geological plan on Map III - C has to be considered to understand the problem: cc' is a shear diatreme with 2 to 6 inches of soft muddy gouge and a sub-vertical dip; mineralization adjacent to this accident pinched out completely on 5320 elevation.

BB' is a sulphide vein with a very sharp contact to the south-west; its dip is sub-vertical and the mineralized width varies from over a foot to the north west down to 1-2 inches to the south east.

These two accidents CC' and BB' outline a heavily shattered zone with three sets of joints:

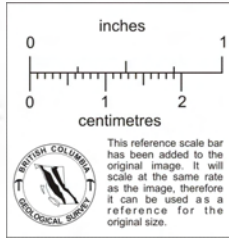
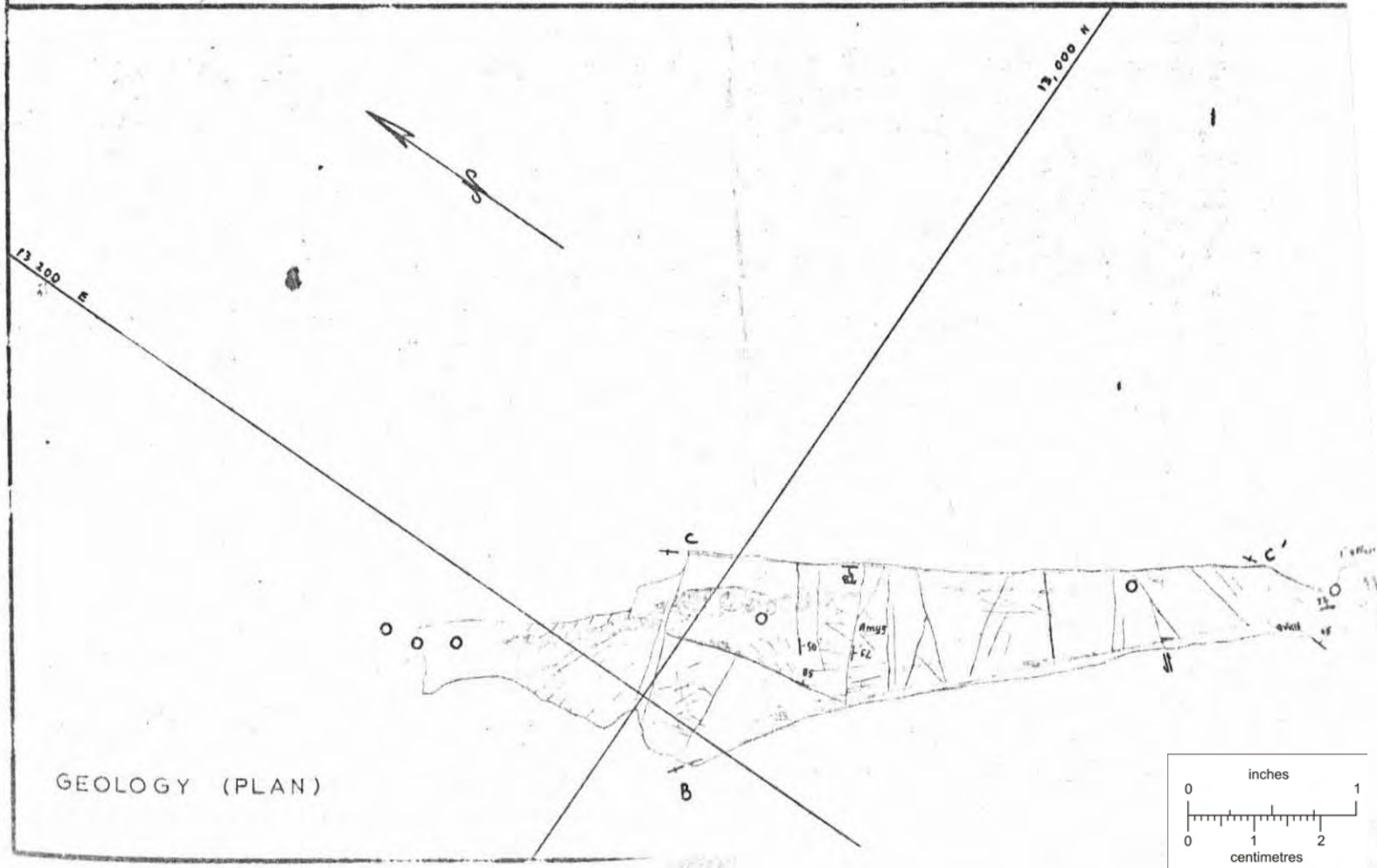
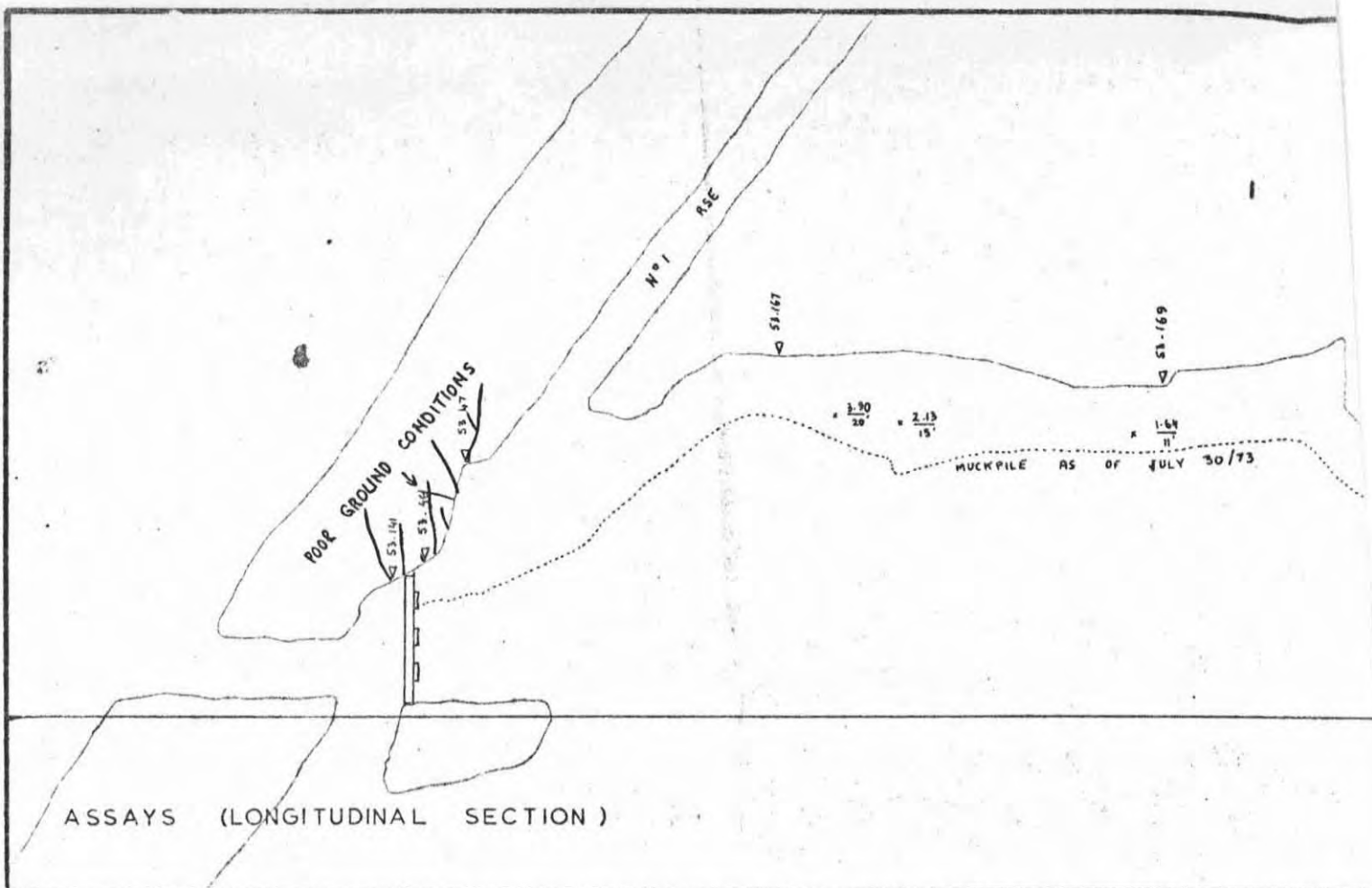
- N 45 - 60 degrees East, 50 to 70 degrees south
- N 0 - 10 degrees East, vertical
- Flat joints.

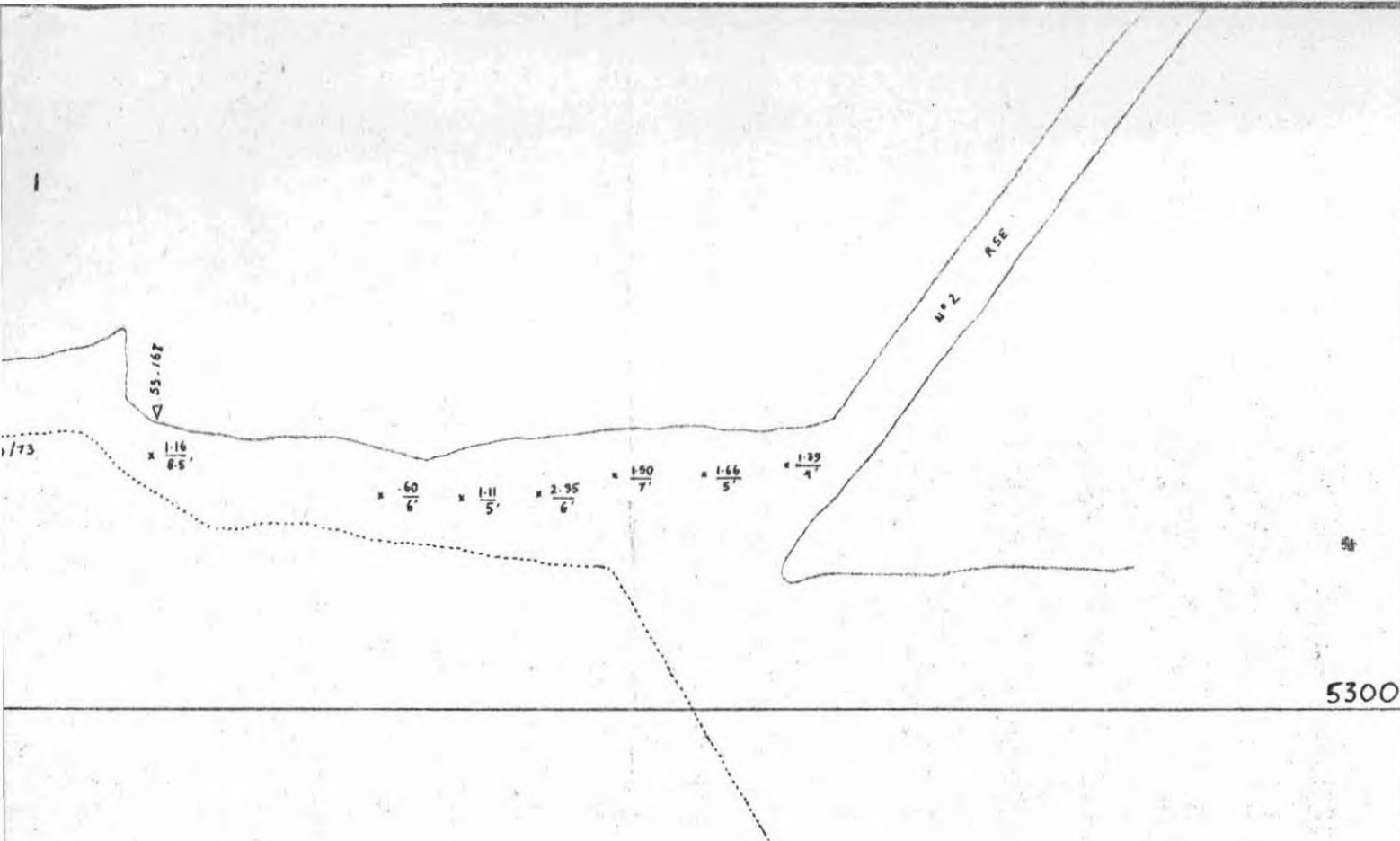
In addition, a limited number of cross-fractures (N 10-20 degrees East, vertical) are present. The mineralization is contained in this zone, with the richest part (Massive and breccia ore) located about 4 feet from the CC' fault.

91 Below 5320, the mineralization spread from CC' to BB', with an average grade generally over 1.80% Cu. The mining procedure was therefore very easy: blast holes were drilled approximately 76 inches inside the two contacts, and the ore would naturally peel off, reducing dilution to nothing and leaving a strong smooth wall.

Above 5320, the economic mineralization being concentrated in the central part of the shear zone, it became quickly impossible to mine from contact to contact without introducing a heavy dilution (30% and over). Attempts were made to cut down the mining width, but due to the discontinuity of the rock cohesion caused by the two diatremes BB' and CC', sloughing in occurred: several large barren fragments came off the walls, diluting the ore, creating hang-ups, hazards and endangering the mining crews.

In addition, minor cave-in occurred in the northern pillar between No. 1 Raise and the empty C longhole stope.





# JORDAN RIVER MINES LTD.

SUNRO MINE

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MAP III - C - RIVER C

SHRINKAGE

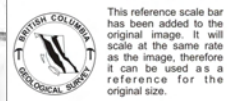
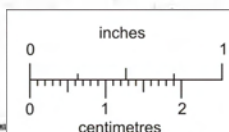
STOPE

DRAWN G. MEUSY

SCALE 1" = 20 Ft

APPROVED

DATE Aug. 2nd /73



This reference scale bar has been added to the original image. It will scale at the same rate as the image, therefore it can be used as a reference for the original size.

In the south-eastern part of the zone (past No. 2 Raise), the sulphide vein allowed a mining width of only 4 feet and under, which has proved impossible to maintain.

Stoping was therefor discontinued and the stope is being drawn empty.

The draw of the southern portion began in July with 1,286 Tons at 1.40% Cu (forecast was 2,221 Tons at 1.52). The draw of the remaining muck should yield a better grade, since reserves are estimated at:

Broken ore - 12,441 Tons at 1.88

#### IV - DIAMOND DRILLING

|               |   |                     |
|---------------|---|---------------------|
| Total Footage | - | 715                 |
| Holes         | - | U.690, U.699, U.702 |
| Recovery      | - | 92%                 |

Shortage in manpower (one driller left this month), and mechanical problems (valve burnt on JV 1, necessary overhaul of both pumps) account for the poor performance in July.

The exploration drilling program has been completed nevertheless; the north-western extension of the Cave Zone has been tested to about 150 feet past the last known informations.

Results establish that there is no economic mineralization along the contact volcanics-intrusive to the north-west.

Very low grade and erratic mineralization, fairly rich in Pyrrhotite has been cored on strike of the Cave A to the north-west of the main haulage, confirming the fact that this Zone is devoid of interest for mining purpose.

Definition drilling has been resumed on 54, from the upramp, to outline and grade the Cave A Zone on Section 1100 (See Map I).

#### V - MISCELLANEOUS

The core of the diamond drill hole IS - 1 drilled at -58 degrees on the claims of R. J. Syndicate has been examined on July 18th/73.

The hole went through a pile of basalt (Metchosin Volcanics) displaying an abnormally high amount of Epidote and Pyrite. Several fault zones were cut, all of them mineralized in Pyrite. Except in the fault zones where Epidote - Calcite - Argillite are present, no hydrothermal alteration has been seen.

The core shows only a few sections of diabase and none of the characters of persistent mineralization at depth.



VI - MID-YEAR ORE RESERVES CALCULATION

Ore reserves in the mine have been blocked out at the end of June 1973 and re-calculated with the same basic principles as in previous estimates.

For cut-off grade calculation, the following factors have been used, based on the milling figures from April 1972 to April 1973:

- Metal recovery equals 93%
- Concentrate Grade equals 25% Cu.
- Ocean shipping, smelting, refining and marketing charges estimated at 20¢ per pound of copper produced.
- Mining, milling, overhead and royalties costs estimated at 32¢ per pound of copper produced (These last two figures were communicated).
- LME Copper price was blocked at 70¢/lb. (which is more than offset by today's price).

The cut-off grades computed from this data are:

.78% Cu. for ore zones wider than 12 Feet  
1.20% Cu. for ore zones where 5 Ft ≤ width ≤ 12 Ft.

The results of the calculation is presented in a tabulated form on the accompanying chart.

In addition, a closer estimation of the possible reserves has been carried out, taking in account for the River Zone the recent results of the exploration drilling program.

Possible ore in Cave Zone between 54 and 57: 160, 400 Tons @ 1.07

Possible ore in River Zone below 51 Level: 156,032 Tons @ 1.48

All the other figures on possible ore given in my 1972 Annual Report remain unchanged.

Analysis of the Ore Reserves Variations:

|                            |              |          |
|----------------------------|--------------|----------|
| Reserves in June 1973      | - 46,439,507 | Lbs. Cu. |
| + Mined out Jan - June     | - 2,816,572  | Lbs. Cu. |
|                            | 49,256,079   | Lbs. Cu. |
| - Reserves in January 1973 | - 48,655,843 | Lbs. Cu. |
| = NET CHANGE               | = + 600,236  | Lbs. Cu. |

This means that the equivalent of 24,600 Tons @ 1.22 of new ore has been found and outlined during the first five months of the year on top of the tonnage equivalent to the production during the same period.

GM/ic  
August 3/73

G. Meusy,  
Mine Geologist.



## SUNRO MINE

## ORE RESE

| ZONE & BLOCK       |      | HEIGHT | LENGTH | BROKEN |        |           | PRO |
|--------------------|------|--------|--------|--------|--------|-----------|-----|
| Name               | Feet | Feet   | TONS   | % Cu   | T x G  | TONS      | %   |
| CAVE A             | 600  | 800    | 5,450  | .91    | 4,959  | 542,705   | 1   |
| CAVE B             | 400  | 500    | 5,705  | 1.04   | 5,933  | 61,955    |     |
| TOTAL CAVE         |      |        | 11,155 | .98    | 10,892 | 604,660   |     |
| RIVER A 1          | 470  | 100    | NIL    | ---    | ---    | 10,220    |     |
| RIVER A 2          | 400  | 200    | NIL    | ---    | ---    | 120,998   |     |
| RIVER A 3          | 270  | 75     | NIL    | ---    | ---    | 10,233    |     |
| TOTAL RIVER A      |      |        | NIL    | ---    | ---    | 141,451   |     |
| B SOUT. PSE PILLAR | 470  | 50     | NIL    | ---    | ---    | 91,900    |     |
| B SILL PILLAR      | 400  | 100    | NIL    | ---    | ---    | 30,100    |     |
| LOWER B            | 350  | 300    | NIL    | ---    | ---    | 200,000   |     |
| B SOUTH            | 200  | 180    | NIL    | ---    | ---    | 10,653    |     |
| B SOUTH-EAST       | 470  | 100    | NIL    | ---    | ---    | 34,191    |     |
| TOTAL RIVER B      |      |        | NIL    | ---    | ---    | 364,844   |     |
| UPPER C            | 700  | 570    | 9,180  | 2.19   | 20,104 | 25,285    |     |
| LOWER C            | 200  | 100    | NIL    | ---    | ---    | NIL       |     |
| TOTAL RIVER C      |      |        | 9,180  | 2.19   | 20,104 | 25,285    |     |
| TOTAL RIVER ZONE   |      |        | 9,180  | 2.19   | 20,104 | 531,580   |     |
| TOTAL CAVE & RIVER |      |        | 20,335 | 1.99   | 20,996 | 1,136,240 |     |
| NEW ZONE           | 150  | 120    | NIL    | ---    | ---    | NIL       |     |
| LAND ZONE          | 200  | 300    | NIL    | ---    | ---    | NIL       |     |
| SUB-TOTALS         |      |        | 20,335 | 1.99   | 20,996 | 1,136,240 |     |

| PROVEN |      |           | PROBABLE |      |         | TOTAL     |      |           |  |
|--------|------|-----------|----------|------|---------|-----------|------|-----------|--|
| S      | % Cu | T x G     | TONS     | % Cu | T x G   | TONS      | % Cu | T x G     |  |
| 05     | 1.18 | 643,170   | 286,145  | 1.21 | 346,068 | 534,300   | 1.19 | 994,197   |  |
| 55     | 1.53 | 95,190    | 28,865   | 1.49 | 43,061  | 96,525    | 1.49 | 144,184   |  |
| 60     | 1.22 | 738,360   | 315,010  | 1.23 | 339,129 | 930,325   | 1.22 | 1,138,381 |  |
| 20     | 2.40 | 24,528    | NIL      | —    | —       | 10,220    | 2.40 | 24,528    |  |
| 08     | 1.32 | 159,997   | 18,762   | 1.37 | 25,766  | 139,760   | 1.33 | 185,763   |  |
| 33     | 1.56 | 15,963    | NIL      | —    | —       | 10,233    | 1.56 | 15,963    |  |
| 51     | 1.42 | 200,488   | 18,762   | 1.37 | 25,766  | 160,213   | 1.41 | 226,254   |  |
| 00     | 1.89 | 173,691   | NIL      | —    | —       | 91,900    | 1.89 | 173,691   |  |
| 00     | 2.25 | 67,725    | NIL      | —    | —       | 30,100    | 2.25 | 67,725    |  |
| 00     | 1.84 | 368,000   | NIL      | —    | —       | 200,000   | 1.84 | 368,000   |  |
| 53     | 2.34 | 43,726    | 18,192   | 2.35 | 42,935  | 26,245    | 2.35 | 86,561    |  |
| 91     | 1.40 | 38,705    | NIL      | —    | —       | 24,191    | 1.40 | 38,705    |  |
| 44     | 1.90 | 691,947   | 18,192   | 2.35 | 42,935  | 383,026   | 1.92 | 734,682   |  |
| 65     | 1.83 | 46,271    | 30,101   | 1.15 | 34,616  | 64,566    | 1.56 | 100,991   |  |
|        | —    | —         | 32,500   | 1.76 | 57,200  | 32,500    | 1.76 | 57,200    |  |
| 85     | 1.23 | 46,271    | 62,401   | 1.47 | 91,816  | 97,046    | 1.43 | 148,191   |  |
| 80     | 1.76 | 938,606   | 29,555   | 1.41 | 260,117 | 640,315   | 1.75 | 1,119,127 |  |
| 240    | 1.17 | 1,676,946 | 414,565  | 1.32 | 349,846 | 1,571,140 | 1.14 | 2,257,502 |  |
|        | —    | —         | 23,844   | 1.67 | 43,320  | 23,844    | 1.67 | 43,320    |  |
|        | —    | —         | 26,775   | 1.10 | 29,152  | 26,775    | 1.10 | 29,152    |  |
| 240    | 1.17 | 1,676,946 | 447,186  | 1.33 | 622,319 | 1,643,759 | 1.13 | 2,130,280 |  |

240 = 1.15 = 1,918,553 Tons 1.22 Calculated by : J. M. M. M.

COMPARISON OF FORECAST TONNAGES AND GRADES (CALCULATED WITH DIAMOND DRILLING INFORMATION) AND ACTUAL TONNAGES (MEASURED) AND GRADES (ESTIMATED WITH MUCK OR CHIP SAMPLING RESULTS) FOR JULY 1973.

| WORKING PLACE             | FORECAST        |       | ACTUAL      |              | REMARKS                                        |
|---------------------------|-----------------|-------|-------------|--------------|------------------------------------------------|
|                           | Tonnage         | Grade | Tonnage     | Grade        |                                                |
| <u>DEVELOPMENT</u>        |                 |       |             |              |                                                |
| 52 Cave A Rse             |                 |       | 21          | .78          | TO COB 16                                      |
|                           |                 |       |             |              |                                                |
|                           |                 |       |             |              |                                                |
|                           |                 |       |             |              |                                                |
|                           |                 |       |             |              |                                                |
|                           |                 |       |             |              |                                                |
|                           |                 |       |             |              |                                                |
|                           |                 |       |             |              |                                                |
| <u>STOPPING</u>           |                 |       |             |              |                                                |
| CAVE A<br>(LINES 12 → 27) | 31 116<br>29560 | .95   | 20,263      | .83<br>47    | Allowed for 10% dilution<br>on forecast 16 812 |
| CAVE B                    | 950<br>807      | .85   | 1,808       | .87<br>42    | 1573                                           |
| RIVER C (Swell)           | 1,260<br>2495   | 1.98  | INCLUDED IN | FIGURE BELOW |                                                |
| RIVER C South             | 2,221<br>3376   | 1.52  | 1,286       | 1.40<br>50   | 1800                                           |
| <u>WASTE</u>              |                 |       |             |              |                                                |
| 53-54 Ramp                |                 |       | 893         | WASTE        | IN C STOPPE                                    |
| RIVER B DP                |                 |       | 80          | .10          | TO COB 8                                       |
| 52-53 MUCK RAISE DRAW     |                 |       | 100         | .10          | TO COB 10                                      |
|                           |                 |       |             |              |                                                |
|                           |                 |       |             |              |                                                |
|                           |                 |       |             |              |                                                |
|                           |                 |       |             |              |                                                |
|                           | 35,547          | 1.02  | 23,558      | .86          | MILLED: 20 225<br>25,671 T @ .838              |



# DIAMOND DRILLING SUMMARY

715-F

- July - 73

| HOLE                    | FROM | TO  | TOTAL | LOCATION                            | CODE           | MACHINE | REMARKS                                         |
|-------------------------|------|-----|-------|-------------------------------------|----------------|---------|-------------------------------------------------|
| JUNE 28/73 - July 12/73 |      |     |       |                                     |                |         |                                                 |
| U-702                   | 517  | 754 | 237   | 51 HANNA ZONE<br>X CUT              | 01.01.01       | JVI     | TOTAL FOOTAGE<br>JUNE 28/73<br>To<br>July 12/73 |
| U-699                   | 0    | 102 | 102   | 51 HANNA ZONE<br>X CUT              | 01.01.01       | JVI     | 339 FT.                                         |
| TOTAL                   |      |     | 339   |                                     | 01.01.01       |         |                                                 |
|                         |      |     |       | COST<br>CODE →                      | 010101         | 100%    |                                                 |
| JULY 13/73 - JULY 30/73 |      |     |       |                                     |                |         |                                                 |
| U-699                   | 102  | 438 | 336   | 51 HANNA<br>ZONE X CUT              | 01.01.01       | JVI     | TOTAL<br>FOOTAGE<br>JULY 13/73<br>To            |
| U-690                   | 0    | 40  | 40    | 54-1100<br>RAMP                     | 01.01.01       | JVI     | JULY 30/73<br>376 FT.                           |
| TOTAL                   |      |     | 376   |                                     | 01.01.01       |         |                                                 |
|                         |      |     |       | FOR MONTH<br>COST CODE:<br>01.01.01 | 715 FT. - 100% |         |                                                 |

FIRST PERIOD: JUNE 28/73 - JULY 12/73 TOTAL FOOTAGE: 339

| DRILLER   | FOOTAGE | TOTAL HRS | COMP. | CONT. | BONUS  | FT/HR | FT/SHIFT | FT/CONT. HR | BONUS<br>SHIFT |
|-----------|---------|-----------|-------|-------|--------|-------|----------|-------------|----------------|
| ANDERSON  | 73      | 24        | 12    | 12    | 65.65  | 3.04  | 24.33    | 6.08        | 21.88          |
| DIDYK     | 48      | 26        | 14.5  | 11.5  | 40.55  | 1.85  | 14.77    | 4.17        | 12.48          |
| GWILLIAM  | 218     | 80        | 25    | 55    | 168.00 | 2.73  | 21.80    | 3.96        | 16.80          |
|           |         |           |       |       |        |       |          |             |                |
|           |         |           |       |       |        |       |          |             |                |
|           |         |           |       |       |        |       |          |             |                |
|           |         |           |       |       |        |       |          |             |                |
|           |         |           |       |       |        |       |          |             |                |
| TOT + AV. | 339     | 130       | 51.5  | 78.5  | 274.20 | 2.54  | 20.30    | 4.74        | 17.05          |

## DIRECT COST:

|                        | TOTAL  | PER FT |
|------------------------|--------|--------|
| LABOUR @ 4.42/HR x 130 | 574.60 | 1.69   |
| BONUS                  | 274.20 | .81    |
| DIAMOND                |        | .38    |
| CORE BOXES             |        | .01    |
| SLUDGE SAMPLING        |        |        |
| MISC.                  |        |        |
| <b>TOTAL:</b>          |        | 2.89   |

## COMPANY TIME:

|                                                      |      |                           |
|------------------------------------------------------|------|---------------------------|
| MOVING / SET UP                                      | 3    | 6%                        |
| REPAIRS                                              | 23.5 | 46%                       |
| NO AIR OR WATER                                      | 1.0  | NO AIR 2%                 |
| PIPE FITTING<br>FREEZING UP<br>SURVEY or SUPERVISION | 12.0 | 23%                       |
| MISC                                                 | 12.0 | 23%                       |
| <b>TOTAL</b>                                         | 51.5 | 37.6%<br>OF TOTAL<br>TIME |

SECOND PERIOD: JULY 15/73 - JULY 30/73 TOTAL FOOTAGE: 376

| DRILLER   | FOOTAGE | TOTAL HRS. | COMP | CONTR. | BONUS  | FT/HR | FT/SHIFT | FT/CONT. HR | BONUS<br>SHIFT |
|-----------|---------|------------|------|--------|--------|-------|----------|-------------|----------------|
| GWILLIAM  | 376     | 96         | 42   | 54     | 242.15 | 3.92  | 31.33    | 6.96        | 20.18          |
|           |         |            |      |        |        |       |          |             |                |
|           |         |            |      |        |        |       |          |             |                |
|           |         |            |      |        |        |       |          |             |                |
|           |         |            |      |        |        |       |          |             |                |
|           |         |            |      |        |        |       |          |             |                |
|           |         |            |      |        |        |       |          |             |                |
|           |         |            |      |        |        |       |          |             |                |
|           |         |            |      |        |        |       |          |             |                |
| TOT + AV. | 376     | 96         | 42   | 54     | 242.15 | 3.92  | 31.33    | 6.96        | 20.18          |

## DIRECT COST:

|                       | TOTAL  | PER FOOT |
|-----------------------|--------|----------|
| LABOUR @ 4.42/HR x 96 | 424.32 | 1.13     |
| BONUS                 | 243.48 | .65      |
| DIAMOND               |        | .38      |
| CORE BOXES            |        | .01      |
| SLUDGE SAMPLING       |        |          |
| MISC.                 |        |          |
| <b>TOTAL :</b>        |        | 2.18     |

## COMPANY TIME:

|                         |    |                            |
|-------------------------|----|----------------------------|
| MOVING / SET UP         | 15 | 36%                        |
| REPAIRS                 | 11 | 26%                        |
| NO AIR AND WATER        | 4  | 10%                        |
| <del>PIPE FITTING</del> | 7  | 17% STRAIGHTEN RODS        |
| SURVEY or SUPERVISION   |    |                            |
| MISC. MOVE GEAR TO SHOP | 5  | 11%                        |
| <b>TOTAL :</b>          | 42 | 43.75%<br>OF TOTAL<br>TIME |

# COWICHAN COPPER CO. LTD. (N.P.L.)

MINE OFFICE:  
RIVER JORDON, B. C.

92C/8E

92C-73

July 31, 1962

3110

Mr. Neil McKechnie,  
Department of Mines,  
Parliament Bldgs.,  
Victoria, B.C.

|                                           |  |  |
|-------------------------------------------|--|--|
| DEPT. OF MINES<br>AND PETROLEUM RESOURCES |  |  |
| Rec'd AUG 2 1962                          |  |  |
| HS                                        |  |  |
|                                           |  |  |

Dear Neil,

As yet we have not washed down the old adits, but I have not forgotten them. The problem is to find someone to do it.

Enclosed is a sketch of 5300 level and the diamond drill holes showing the dykes. This is the level which you mapped a couple of months ago. I have marked all the intersections of dyke and indicated what, to me, seems the most logical way to join them. The result is an amazing sinuous, serpentine dyke. Since these configurations have no apparent effect on either the faults or the ore they must be pre-ore. NDARK

I very much doubt, that the dyke has been folded into this shape, or that it is following an older structure in the basalts. The basalts, except in the immediate vicinity of the ore, are hard, massive and fresh in appearance. There is no schistosity nor metamorphism, which one would normally find with such tight and complex folding. Then, what caused this? -

I am wondering if perhaps the dykes are not intrusive, but are part of the flows and that this peculiar configuration is a flow structure developed during the final stage of advance when the flow was viscous. This seems more probable and could explain why the dyke is so difficult to differentiate from the basalts.

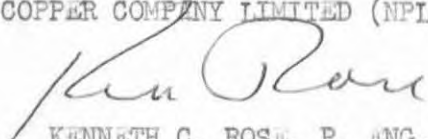
To those who might argue that this peculiar structure occurs only in the ore zone and hence has a bearing on the ore deposition, I would point out, that it is only in the ore zone, that sufficient close drilling has been done to delineate the structure. This structure probably extends into the wall rocks, but has not been traced.

What are your thoughts about this? Have you encountered any comparable structures? I would like to hear your suggestions as well as those of any one else whom you can interest.

Best Regards

COWICHAN COPPER COMPANY LIMITED (NPL)

PROPERTY FILE

  
KENNETH C. ROSE, P. ENG.